PH 530

Light–Matter Interaction

Lecture Notes 2026

Prof. Anshuman Kumar
Department of Physics, IIT Bombay

Contents

I Quantization of Light 1
II Coherent States 4
III Squeezed States 7
IV Nonlinear Optics & SPDC 12
V SU(2)/SU(1,1) & Beam Splitters 18
VI Balanced Homodyne Detection 25
VII Quantum Key Distribution 30
VIII Atom–Light Interaction 32

Prof. Anshuman Kumar
Department of Physics
Indian Institute of Technology Bombay

Teaching Assistants
Devansh Satra *(EP'26)*
Yash Gupta *(EP'27)*

Academic Year 2025–26

Part I: Quantization of Light

From classical harmonic oscillators to quantum field modes

1 Classical Harmonic Oscillator

The harmonic oscillator is the fundamental building block of quantum optics. Every mode of the EM field behaves as an independent QHO.

The classical equation of motion is

$$\ddot{x} = -\omega^2 x.$$

A general solution is

$$x(t) = x_0 \sin(\omega t + \phi),$$

with momentum

$$p(t) = m\omega x_0 \cos(\omega t + \phi).$$

The total energy is

$$E = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}\left(\omega^2 x^2 + p^2\right).$$

2 Light in a 1D Cavity

Metallic boundary conditions force $E_x(0,t) = 0$, selecting a discrete spectrum of standing-wave modes $k = n\pi/L$.

Consider light confined in a one-dimensional cavity of length L, polarized along $\hat{x}$ and propagating along $\hat{z}$. The electric field is written as

$$E_x(z,t) = E_{01}\cos(kz - \omega t + \phi_1) + E_{02}\cos(kz + \omega t + \phi_2).$$

The boundary condition $E_x(0,t) = 0$ gives

$$E_{01} = -E_{02} \equiv \frac{E_0}{2}, \qquad \phi_1 = -\phi_2 \equiv \phi,$$

hence

$$E_x(z,t) = E_0 \sin(kz)\sin(\omega t - \phi).$$

From $\nabla \times \boldsymbol{E} = -\partial_t \boldsymbol{B}$,

$$B_y(z,t) = \frac{E_0}{c}\cos(kz)\cos(\omega t - \phi),$$

with dispersion relation $\omega = ck$.

3 Energy of the Electromagnetic Field

The total energy stored in the field is

$$H = \int A\,dz \left(\frac{\epsilon_0 E^2}{2} + \frac{B^2}{2\mu_0} \right).$$

The electric and magnetic contributions alternate in time, just like kinetic and potential energy in a mechanical oscillator. Their sum is constant.

Substituting the fields and integrating over the cavity,

$$H = \frac{1}{2}\epsilon_0 A E_0^2 \left[\frac{1}{2}\sin^2(\omega t - \phi) + \frac{1}{2}\cos^2(\omega t - \phi) \right] \frac{L}{2}.$$

Thus the energy oscillates between electric and magnetic contributions. Choosing $\phi = 0$,

$$H = \frac{1}{2}\left(\omega^2 q^2 + p^2\right),$$

with canonical variables

$$p(t) = \left(\frac{\epsilon_0 V}{2}\right)^{1/2} E_0 \cos\omega t,$$
$$q(t) = \left(\frac{\epsilon_0 V}{2\omega^2}\right)^{1/2} E_0 \sin\omega t.$$

Light as a Harmonic Oscillator

The electromagnetic field in a cavity is *mathematically identical* to a harmonic oscillator. Quantizing the field is therefore equivalent to quantizing the harmonic oscillator.

4 Quadratures

For $\phi \neq 0$,

$$\begin{aligned} E_x(z,t) &= E_0 \sin kz(\sin\omega t\cos\phi - \cos\omega t\sin\phi) \\ &= E_1 \sin\omega t + E_2\cos\omega t, \end{aligned}$$

where

$$E_1 = E_0 \sin kz \cos\phi, \qquad E_2 = -E_0 \sin kz \sin\phi.$$

Uncertainty relation

$$\Delta X_1\,\Delta X_2 \geq \frac{1}{4}$$

5 Phasor Representation

The field may be written as $E(z,t) = E_0(z)e^{-i\omega t}$ with complex $E_0(z)$. Switching to dimensionless variables via the single-photon field amplitude

$$E_{0N} = \sqrt{\frac{4\hbar\omega}{\epsilon_0 V}},$$

we write

$$E_x(z,t) = \sqrt{\frac{4\hbar\omega}{\epsilon_0 V}}\sin(kz)\left[\hat{X}_1(t)\cos\phi + \hat{X}_2(t)\sin\phi\right].$$

Define the quadrature operators

$$\hat{X}_1 = \left(\frac{\omega}{2\hbar}\right)^{1/2} \hat{q}, \qquad \hat{X}_2 = \left(\frac{1}{2\hbar\omega}\right)^{1/2} \hat{p}.$$

From $[\hat{q}, \hat{p}] = i\hbar$,

$$[\hat{X}_1, \hat{X}_2] = \frac{i}{2} \quad \Longrightarrow \quad \Delta X_1 \, \Delta X_2 \geq \frac{1}{4}.$$

6 Creation and Annihilation Operators

For $\phi = 0$,

$$\hat{a} = \hat{X}_1 + i\hat{X}_2, \qquad \hat{a}^\dagger = \hat{X}_1 - i\hat{X}_2.$$

Commutation

$[\hat{a}, \hat{a}^\dagger] = 1$

Inverting,

$$\hat{X}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}, \qquad \hat{X}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}.$$

The Hamiltonian becomes

$$\hat{H} = \hbar\omega(\hat{X}_1^2 + \hat{X}_2^2) = \frac{\hbar\omega}{2}(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}).$$

Since $\hat{X}_1, \hat{X}_2$ are Hermitian, $[\hat{a}, \hat{a}^\dagger] = 1$, and

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).$$

Time evolution: $\hat{a}(t) = \hat{a}(0)e^{-i\omega t}$.

7 Number States

Define the number operator $\hat{N} = \hat{a}^\dagger\hat{a}$. Its eigenstates satisfy

$$\hat{N}\,|n\rangle = n\,|n\rangle\,,$$

Fock states have definite photon number but completely undefined phase. They form a complete orthonormal basis for the single-mode Hilbert space.

with ladder actions

$$\hat{a}\,|n\rangle = \sqrt{n}\;|n-1\rangle\,,$$
$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\;|n+1\rangle\,.$$

Expectation values:

$$\langle\hat{a}(t)\rangle_n = 0, \qquad \langle(\Delta X_1)^2\rangle = \frac{1}{4}(2n+1).$$

Thus number states have zero mean field but large fluctuations.

Part II: Coherent States

The most classical quantum states of light

8 Coherent States

Define coherent states as eigenstates of $\hat{a}$:

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle\,.$$

Coherent states were introduced by Glauber (1963) to describe laser output. They are the closest quantum analogue of a classical monochromatic wave.

Expanding $|\alpha\rangle = \sum_n c_n\,|n\rangle$ and applying $\hat{a}$,

$$c_{n+1} = \frac{\alpha}{\sqrt{n+1}}c_n \quad\Longrightarrow\quad c_n = \frac{\alpha^n}{\sqrt{n!}}c_0.$$

Normalization gives $c_0 = e^{-|\alpha|^2/2}$, so

$$|\alpha\rangle = \sum_{n=0}^{\infty} e^{-|\alpha|^2/2}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,.$$

9 Overlap of Coherent States

$$\langle\alpha|\beta|\alpha|\beta\rangle = \exp\left(-\frac{|\alpha|^2+|\beta|^2-2\alpha^*\beta}{2}\right) \neq 0.$$

Thus coherent states are *not* orthogonal.

10 Resolution of Identity

$$\mathbb{I} = \int \frac{d^2\alpha}{\pi}\,|\alpha\rangle\,\langle\alpha|\,.$$

Proof proceeds by inserting number states and using

$$\int_0^{2\pi} d\theta\, e^{i(m-n)\theta} = 2\pi\delta_{mn}.$$

Proof uses $\int_0^{2\pi} d\theta\, e^{i(m-n)\theta} = 2\pi\delta_{mn}$ after switching to polar coordinates $\alpha = re^{i\theta}$.

Hence the coherent-state basis is **overcomplete**.

11 Coherent State Statistics

Using $\hat{a}(t) = \hat{a}(0)e^{-i\omega t}$, the quadrature expectation values are

$$\langle\alpha|\hat{X}_1(t)|\alpha\rangle = \mathrm{Re}(\alpha e^{-i\omega t}),$$
$$\langle\alpha|\hat{X}_2(t)|\alpha\rangle = \mathrm{Im}(\alpha e^{-i\omega t}).$$

The variance of each quadrature is

$$\langle\alpha|(\Delta\hat{X}_i)^2|\alpha\rangle = \frac{1}{4}, \qquad i = 1, 2.$$

11.1 Signal-to-Noise Ratio

The SNR is defined as

$$\text{SNR} = \frac{\langle\hat{X}_1\rangle^2}{\langle(\Delta\hat{X}_1)^2\rangle}.$$

The SNR grows as $\sqrt{N}$ with photon number, explaining why laser beams behave increasingly classically at high power.

For large $|\alpha|$,

$$\frac{\langle\hat{X}_1\rangle}{\sqrt{\langle(\Delta\hat{X}_1)^2\rangle}} \xrightarrow{|\alpha|\to\infty} \sqrt{2}\,\text{Re}(\alpha e^{-i\omega t}).$$

Thus for large $|\alpha|$, coherent states behave classically with increasing SNR.

12 Minimum Uncertainty Property

Minimum Uncertainty State

For any state: $\langle(\Delta\hat{X}_1)^2\rangle\langle(\Delta\hat{X}_2)^2\rangle \geq 1/16$.
For coherent states: $\langle(\Delta\hat{X}_1)^2\rangle = \langle(\Delta\hat{X}_2)^2\rangle = 1/4$, so

$$\langle(\Delta\hat{X}_1)^2\rangle\langle(\Delta\hat{X}_2)^2\rangle = \frac{1}{16}.$$

Coherent states are **minimum uncertainty states** with equal quadrature variances.

13 Number Statistics of Coherent States

$$\langle\alpha|\hat{N}|\alpha\rangle = |\alpha|^2 \equiv \bar{n}, \qquad \langle(\Delta\hat{N})^2\rangle = \bar{n}.$$

Variance equals mean — the hallmark of a Poisson distribution.

Poisson statistics

$\text{Var}(\hat{N}) = \langle\hat{N}\rangle = \bar{n}$

13.1 Poisson Distribution

The probability of measuring n photons is

$$\Pr(N = n) = |\langle n|\alpha|n|\alpha\rangle|^2 = \frac{\bar{n}^n}{n!}e^{-\bar{n}},$$

a Poisson distribution with mean $\bar{n} = |\alpha|^2$.

14 Phase-Space Picture of Coherent States

The Hamiltonian may be written as

$$\hat{H} = \hbar\omega(\hat{X}_1^2 + \hat{X}_2^2) = \hbar\omega\left(\hat{N} + \frac{1}{2}\right).$$

Physical insight

The phase-space (Wigner function) picture provides the most intuitive way to visualise quantum states of light. A coherent state is a Gaussian blob rotating at the optical frequency.

A coherent state $|\alpha\rangle$ corresponds to a localized Gaussian distribution in phase space, centered at $(a_1, a_2) = (\operatorname{Re}\alpha, \operatorname{Im}\alpha)$, with radius $\sqrt{\bar{n} + 1/2}$.

The electric field expectation value oscillates as

$$\langle \hat{E}(t) \rangle \propto \langle \hat{X}_1 \rangle \cos \omega t + \langle \hat{X}_2 \rangle \sin \omega t,$$

producing a well-defined classical waveform with small fluctuations.

15 Phase-Space Picture of Number States

For a Fock state $|n\rangle$,

$$\hat{H} = \hbar\omega(\hat{X}_1^2 + \hat{X}_2^2) = \hbar\omega\left(n + \frac{1}{2}\right).$$

In phase space, the number state corresponds to a *ring* of radius $\sqrt{n + 1/2}$. Since the phase is completely undefined, the probability distribution is rotationally symmetric, and $\langle n|\hat{E}(t)|n\rangle = 0$.

16 Intuitive Number–Phase Uncertainty

From the phase-space picture, fluctuations in photon number correspond to radial uncertainty:

$$\Delta n = \left(|\alpha| + \tfrac{1}{2}\right)^2 - \left(|\alpha| - \tfrac{1}{2}\right)^2 = |\alpha| = \sqrt{\bar{n}}.$$

The corresponding phase uncertainty is $\Delta\phi \sim 1/\sqrt{\bar{n}}$, giving

$$\Delta n \cdot \Delta\phi \sim 1.$$

Part III: Squeezed States

Non-classical light with reduced quadrature noise

17 Squeezed States

In a squeezed state, the uncertainty distribution in phase space becomes elliptical. One quadrature has reduced variance at the expense of increased variance in the conjugate quadrature:

$$\Delta a_1^2 < \frac{1}{4}, \quad \Delta a_2^2 > \frac{1}{4} \quad \text{(or vice versa)}.$$

Squeezing was first observed experimentally by Slusher et al. (1985). Modern experiments achieve $> 15\,\text{dB}$ of squeezing, used in Advanced LIGO for gravitational-wave detection.

Depending on the orientation of the ellipse, the state may be **phase-squeezed** or **amplitude-squeezed**.

18 Electric Field Fluctuations

The electric field operator can be written as

$$\hat{E}(t) \propto \hat{X}_1 \cos \omega t + \hat{X}_2 \sin \omega t.$$

Field operator

$\hat{E}(t) \propto \hat{X}_1 \cos \omega t + \hat{X}_2 \sin \omega t$

Hence,

$$\hat{E}^2(t) \propto \hat{X}_1^2 \cos^2 \omega t + \hat{X}_2^2 \sin^2 \omega t + \{\hat{X}_1, \hat{X}_2\} \sin \omega t \cos \omega t.$$

In squeezed states, the noise in $\hat{E}(t)$ becomes phase-dependent, leading to reduced fluctuations in selected quadratures.

19 Vacuum, Coherent Vacuum, and Squeezed Vacuum

- **Vacuum state**: centered at the origin with circular uncertainty.
- **Coherent vacuum**: displaced vacuum state.
- **Squeezed vacuum**: vacuum with elliptical uncertainty.

These transformations preserve phase-space area (Liouville's theorem).

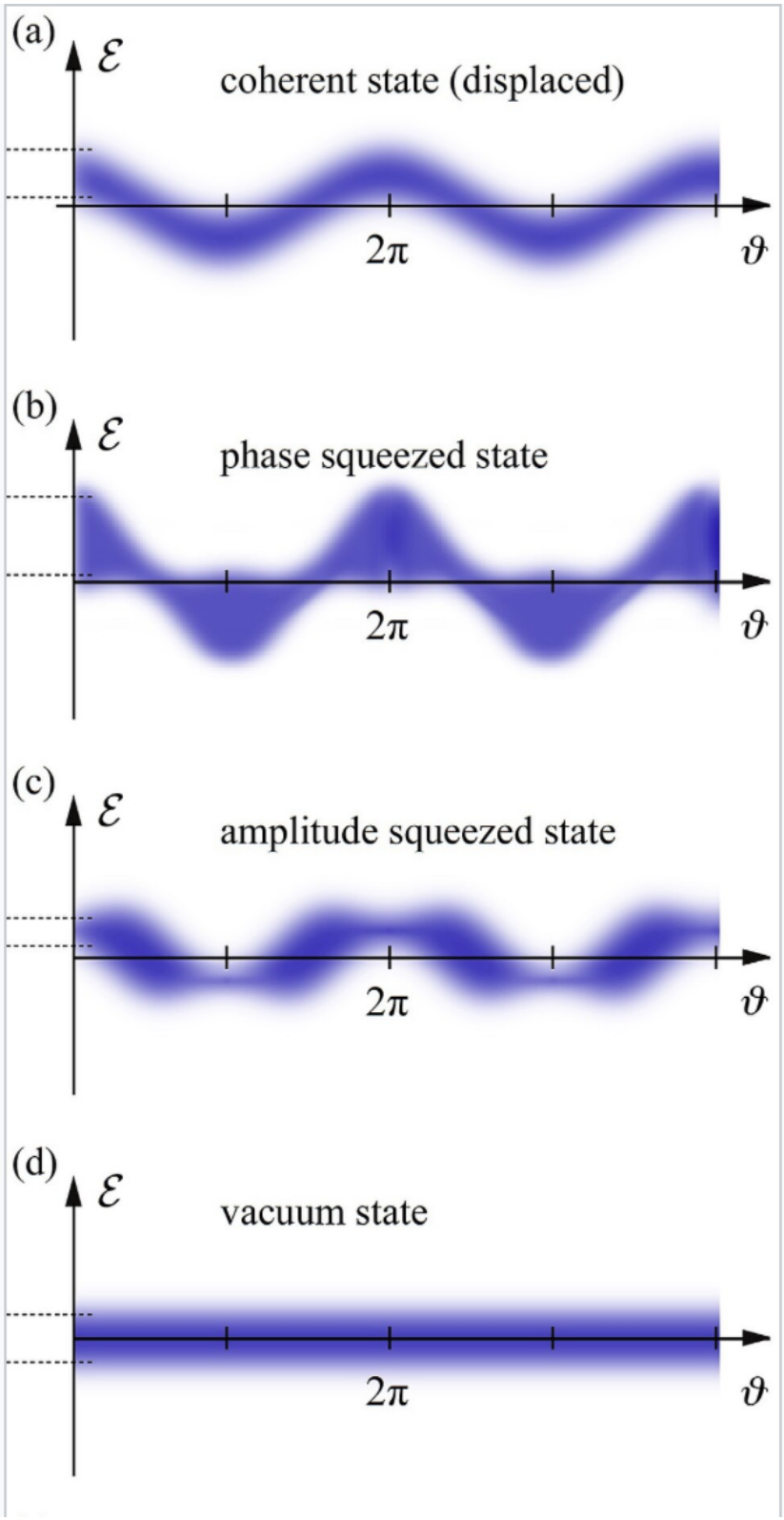

Figure 1: Phase-space representations of several quantum states of light.
Source: Wikipedia.

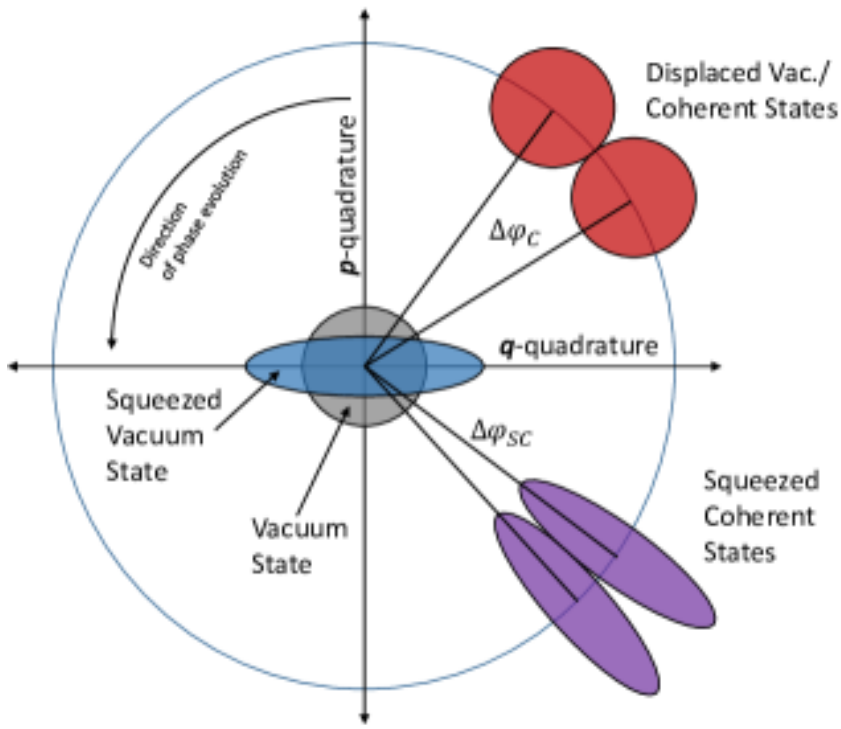

Figure 2: Electric field noise as a function of phase ϑ for (a) coherent, (b) phase-squeezed, (c) amplitude-squeezed, and (d) vacuum states. *Source: Quantum* **5**, *458 (2021).*

20 Symplectic Structure

Quadrature transformations belong to the symplectic group

$$\mathrm{Sp}(2,\mathbb{R}) \cong \mathrm{SL}(2,\mathbb{R}),$$

which preserves the commutation relation $[\hat{X}_1, \hat{X}_2] = i/2$.

Let $\boldsymbol{R} = (\hat{X}_1, \hat{X}_2)^T$. Under a linear transformation $\boldsymbol{R}' = S\boldsymbol{R}$,

$$[R'_i, R'_j] = (S^T \Omega S)_{ij}, \quad \Omega = \frac{i}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Preservation of commutation relations requires $S^T \Omega S = \Omega$; such matrices are **symplectic**.

$\mathrm{Sp}(2,\mathbb{R}) \cong \mathrm{SL}(2,\mathbb{R})$ is the group of 2×2 real matrices with determinant 1. It has three generators: two squeezings and one rotation.

20.1 Example: Rotation

$$S = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

One verifies $S^T \Omega S = \Omega$, hence rotations are valid symplectic transformations.

20.2 Example: Anisotropic Scaling (Squeezing)

Let $\hat{X}_1' = \alpha \hat{X}_1$, $\hat{X}_2' = \beta \hat{X}_2$. The symplectic condition requires $\alpha\beta = 1$, so we choose

$$S = \begin{pmatrix} e^{-r} & 0 \\ 0 & e^{r} \end{pmatrix}, \quad \det S = 1.$$

21 Squeezing Operator

Define a unitary operator $\hat{U}(r) = e^{-ir\hat{G}}$ with Hermitian $\hat{G}$. The transformed operator satisfies

$$\frac{d\hat{A}(r)}{dr} = i\hat{U}^\dagger(r)[\hat{G}, \hat{A}]\hat{U}(r).$$

Squeezing

$\hat{S}(r) = \exp\left[\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})\right]$

For the annihilation operator,

$$\hat{a}(r) = \mu(r)\hat{a} + \nu(r)\hat{a}^\dagger.$$

Choosing the generator such that $[\hat{G}, \hat{a}] \sim \hat{a}^\dagger$ leads to

$$\hat{a}(r) = \hat{a}\cosh r - \hat{a}^\dagger \sinh r.$$

22 Construction of the Squeezing Operator

We require the quadrature transformations

$$\hat{U}^\dagger(r)\hat{X}_1\hat{U}(r) = e^{-r}\hat{X}_1,$$
$$\hat{U}^\dagger(r)\hat{X}_2\hat{U}(r) = e^{+r}\hat{X}_2,$$

equivalently $\hat{a}(r) = \hat{a}\cosh r - \hat{a}^\dagger \sinh r$.

This transformation is analogous to a Lorentz boost in Minkowski space, with r playing the role of the rapidity.

23 Generator of Squeezing

The most general Hermitian quadratic generator is

$$\hat{G} = \alpha\hat{a}^{\dagger 2} + \beta\hat{a}^2 + \gamma\hat{a}^\dagger\hat{a} + \delta,$$

with $\beta = \alpha^*$ and $\gamma, \delta \in \mathbb{R}$. Choosing $\alpha = -i/2$, $\beta = i/2$, $\gamma = 0$,

$$\hat{G} = \frac{i}{2}(\hat{a}^2 - \hat{a}^{\dagger 2}).$$

Thus the squeezing operator is

$$\hat{S}(r) = e^{-ir\hat{G}} = \exp\left[\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})\right].$$

In quadrature form, squeezing corresponds to the symplectic matrix

$$S(r) = \begin{pmatrix} e^{-r} & 0 \\ 0 & e^{+r} \end{pmatrix}, \quad \det S = 1.$$

24 Experimental Origin

Squeezed states arise naturally in nonlinear optics, particularly via second-order nonlinear processes ($\chi^{(2)}$) such as parametric down-conversion. The squeezing operator is

$$\hat{S}(\zeta) = \exp\left(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2}\right), \quad \zeta = re^{i\phi}.$$

Physical insight

$\chi^{(2)}$ materials such as BBO, KTP, and $LiNbO_3$ are the standard workhorses for squeezing experiments. They require phase matching to be efficient.

25 Squeezed Vacuum State

The squeezed vacuum is defined as $|0, r\rangle = \hat{S}(r)\,|0\rangle$. Expanding in number basis and using the annihilation condition $\hat{a}(r)\,|0, r\rangle = 0$, one obtains the recursion

$$\cosh r\,\sqrt{n+1}\,C_{n+1} = \sinh r\,\sqrt{n}\,C_{n-1},$$

implying $C_1 = C_3 = C_5 = \cdots = 0$. The normalized squeezed vacuum is

$$|0, r\rangle = \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \frac{\sqrt{(2n)!}}{2^n n!} (\tanh r)^n\,|2n\rangle\,.$$

Note

Only even Fock states appear because the generator $\hat{a}^2$ creates/destroys photons in pairs.

26 Rotation and Squeezing

The general squeezing operator including rotation is

$$\hat{S}(\zeta) = \exp\left[\frac{1}{2}(\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2})\right], \quad \zeta = re^{i\theta}.$$

27 Expectation Values

For a squeezed coherent state $\hat{S}(r)\hat{D}(\alpha)\,|0\rangle$,

$$\langle \hat{n} \rangle = \sinh^2 r + |\alpha|^2,$$
$$\langle (\Delta \hat{n})^2 \rangle \approx |\alpha|^2 e^{2r}, \quad (|\alpha|^2 \gg \sinh^2 r).$$

For the squeezed vacuum ($\alpha = 0$),

$$\langle (\Delta \hat{n})^2 \rangle = 2\bar{n}(\bar{n}+1), \quad \bar{n} = \sinh^2 r.$$

Part IV: Nonlinear Optics and SPDC

Generating entanglement from $\chi^{(2)}$ interactions

28 Implementing Squeezing: Nonlinear Polarization

When light propagates through matter it induces a polarization field. In general,

$$\mathbf{P} = \varepsilon_0 \sum_{n=1}^{\infty} \chi^{(n)} \mathbf{E}^n,$$

where $\chi^{(n)}$ are the nth-order susceptibilities. The linear term gives ordinary refraction; higher-order terms give nonlinear optical effects.

The polarization field acts as a source term in Maxwell's equations, driving new electromagnetic fields at new frequencies.

29 Classical Nonlinear Oscillator Model

An atom modeled as a bound electron in a harmonic potential $V(x) = \frac{1}{2}m\omega_0^2 x^2$ with interaction force $F(t) = -eE(t)$ gives the linear response

$$x^{(1)}(t) = -\frac{eE_0}{m(\omega_0^2 - \omega^2)} e^{-i\omega t},$$

and single-atom dipole moment $p = -ex^{(1)}(t)$. For a medium of density n,

$$P^{(1)} = \frac{ne^2}{m(\omega_0^2 - \omega^2)} E_0 e^{-i\omega t}.$$

30 Origin of Nonlinearity

In reality the atomic potential deviates from purely harmonic:

$$V(x) = \frac{1}{2}m\omega_0^2 x^2 + \alpha x^3 + \beta x^4 + \cdots$$

Expanding $x = x^{(1)} + x^{(2)} + \cdots$ perturbatively,

Physical insight

Centrosymmetric crystals have inversion symmetry $x \to -x$, which forces $\chi^{(2)} = 0$. Hence second-order nonlinearity requires a non-centrosymmetric material.

30.1 Second-Order Response

The second-order equation is $m\ddot{x}^{(2)} + m\omega_0^2 x^{(2)} = -3\alpha (x^{(1)})^2$.

Using $E(t) = E_0 \cos \omega t$:

$$(x^{(1)})^2 \propto E_0^2 \cos^2 \omega t = \frac{E_0^2}{2}(1 + \cos 2\omega t).$$

Thus the second-order displacement contains a DC term and a second-harmonic term:

$$x_{\mathrm{DC}}^{(2)} = -\frac{3\alpha a^2 E_0^2}{2m\omega_0^2},$$
$$x_{2\omega}^{(2)} = -\frac{3\alpha a^2 E_0^2}{2m(\omega_0^2 - 4\omega^2)} e^{-2i\omega t}.$$

The corresponding polarization is $P^{(2)} = nex^{(2)}$.

In centrosymmetric crystals, inversion symmetry implies $\chi^{(2)} = 0$.

31 Three-Wave Mixing

For a multi-frequency field

$$E(t) = \tilde{E}_1 e^{-i\omega_1 t} + \tilde{E}_2 e^{-i\omega_2 t} + \text{c.c.},$$

second-order nonlinearities generate new frequencies:

$$2\omega_1, \quad 2\omega_2, \quad \omega_1 + \omega_2, \quad |\omega_1 - \omega_2|.$$

32 Maxwell Equations with Polarization

The relevant Maxwell equations are

$$\nabla \cdot \mathbf{D} = \rho_{\mathrm{ext}},$$
$$\nabla \cdot \mathbf{B} = 0,$$
$$\nabla \times \mathbf{B} = \mu_0 \frac{\partial \mathbf{D}}{\partial t} + \mu_0 \mathbf{J}_{\mathrm{ext}}.$$

Combining these gives the wave equation

$$\left(\nabla \times \nabla \times + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\mathbf{E} = -\mu_0 \frac{\partial^2 \mathbf{P}}{\partial t^2} - \mu_0 \frac{\partial \mathbf{J}_{\mathrm{ext}}}{\partial t}.$$

Thus polarization acts as a source term for new electromagnetic fields.

33 Materials for Quantum Optics

Second-order nonlinearities are exploited in non-centrosymmetric crystals such as BBO, KTP, and $LiNbO_3$.

34 Origin of the SPDC Hamiltonian

In SPDC, a pump photon at ω_p splits into signal and idler photons:

$$\omega_p = \omega_s + \omega_i.$$

SPDC Hamiltonian

$H_{\mathrm{SPDC}} = i\hbar\kappa(e^{i\phi}\hat{a}_s^\dagger \hat{a}_i^\dagger - \text{h.c.})$

The interaction Hamiltonian is

$$H_{\mathrm{int}} = -\int d^3r\, \mathbf{P}^{(2)} \cdot \mathbf{E} \propto \int d^3r\, \chi^{(2)} E^3.$$

The total electric field in the medium is $\hat{E} = E_p + \hat{E}_s + \hat{E}_i$, where the pump is treated classically. Keeping only energy- and phase-matched terms,

$$H_{\text{SPDC}} = i\hbar\kappa\left(e^{i\phi}\hat{a}_s^\dagger\hat{a}_i^\dagger - e^{-i\phi}\hat{a}_s\hat{a}_i\right).$$

The free Hamiltonian is $H_0 = \hbar\omega_s\hat{a}_s^\dagger\hat{a}_s + \hbar\omega_i\hat{a}_i^\dagger\hat{a}_i$. For perfect phase matching $\Delta = \omega_s + \omega_i - \omega_p = 0$, the interaction-picture Hamiltonian is unchanged in form, and the evolution operator is

$$U(t) = \exp\left[\kappa t\left(e^{i\phi}\hat{a}_s^\dagger\hat{a}_i^\dagger - e^{-i\phi}\hat{a}_s\hat{a}_i\right)\right].$$

Physical Consequences

This interaction generates:

- Two-mode squeezed vacuum
- Entangled photon pairs
- Quantum correlations relevant to interferometry and Hawking radiation analogues

35 Experimental Implementation of Squeezing

SPDC arises from light–matter interaction in a $\chi^{(2)}$ medium. A strong classical pump at ω_p produces signal and idler photons satisfying $\omega_p = \omega_s + \omega_i$, providing an experimental realization of two-mode squeezing, parametric amplification, and entangled photon pairs.

35.1 Degenerate SPDC

In the degenerate case $\omega_s = \omega_i = \omega_p/2$, the SPDC Hamiltonian reduces to a single-mode squeezing Hamiltonian:

$$H_{\text{deg}} = \frac{i\hbar\kappa}{2}\left(e^{i\phi}\hat{a}^{\dagger 2} - e^{-i\phi}\hat{a}^2\right).$$

35.2 Coupled-Mode Description

A more general multimode interaction Hamiltonian is

$$H_I = \hbar\sum_{k_s,k_i} g_{k_s k_i}\left(\hat{a}_{k_s}^\dagger\hat{a}_{k_i}^\dagger e^{-i\Delta kz} + \text{h.c.}\right),$$

with phase mismatch $\Delta k = k_p - k_s - k_i$. Solving the coupled Heisenberg equations yields

$$\hat{a}_{k_s}(t) = \hat{a}_{k_s}(0)\cosh(gt) - i\hat{a}_{k_i}^\dagger(0)\sinh(gt),$$

describing parametric amplification.

35.3 Phase Matching Function

Phase matching

$$|g(\Delta k)|^2 \propto \text{sinc}^2\left(\frac{\Delta kL}{2}\right)$$

$$g_{k_s k_i} \propto \int_0^L dz\, e^{i\Delta k z} = \frac{e^{i\Delta k L} - 1}{i\Delta k},$$

with intensity spectrum

$$|g(\Delta k)|^2 \propto \operatorname{sinc}^2\!\left(\frac{\Delta k L}{2}\right),$$

which peaks sharply at $\Delta k = 0$.

36 Two-Mode Squeezing Formalism

Define the quadrature operators

$$\hat{X} = \frac{\hat{a} + \hat{b} + \hat{a}^\dagger + \hat{b}^\dagger}{2}, \qquad \hat{P} = \frac{\hat{a} + \hat{b} - \hat{a}^\dagger - \hat{b}^\dagger}{2i}.$$

The two-mode squeezing operator is

$$S_2(\xi) = \exp\left(\xi^* \hat{a}\hat{b} - \xi \hat{a}^\dagger \hat{b}^\dagger\right), \quad \xi = r e^{i\theta}.$$

Using the Baker–Campbell–Hausdorff relation,

$$\begin{aligned} S_2^\dagger(\xi)\, \hat{a}\, S_2(\xi) &= \hat{a} \cosh r - e^{i\theta} \hat{b}^\dagger \sinh r, \\ S_2^\dagger(\xi)\, \hat{b}\, S_2(\xi) &= \hat{b} \cosh r - e^{i\theta} \hat{a}^\dagger \sinh r. \end{aligned}$$

The two-mode squeezed vacuum $|\xi\rangle_2 = S_2(\xi)\,|0,0\rangle$ has

$$\langle \hat{X} \rangle = \langle \hat{P} \rangle = 0, \quad \langle \hat{X}^2 \rangle \sim e^{-2r}, \quad \langle \hat{P}^2 \rangle \sim e^{2r}.$$

37 Two-Mode Squeezing: Explicit State

We solve $S_2(\xi)\,|0,0\rangle$ explicitly. Define $\hat{D} = \hat{a}_s^\dagger \hat{a}_s - \hat{a}_i^\dagger \hat{a}_i$; since $[\hat{D}, S_2(\xi)] = 0$ and $\hat{D}\,|0,0\rangle = 0$,

$$S_2(\xi)\,|0,0\rangle \in \operatorname{span}\{|n,n\rangle\}_{n=0}^{\infty}.$$

Physical insight

The state lives entirely in the equal-photon-number subspace $\operatorname{span}\{|n,n\rangle\}$ because the number-difference operator commutes with S_2.

Writing $|\psi\rangle = \sum_n C_n\,|n,n\rangle$ and using the transformed annihilation operators, the annihilation condition gives the recursion $C_{n+1} = e^{i\phi} \tanh r\, C_n$, so

$$\boxed{S_2(\xi)\,|0,0\rangle = \frac{1}{\cosh r} \sum_{n=0}^{\infty} (e^{i\phi} \tanh r)^n\, |n\rangle_s\, |n\rangle_i\,.}$$

38 Polarization-Entangled Photon Sources

38.1 SPDC Sources

SPDC arises from the nonlinear polarization response:

$$\mathbf{P} = \varepsilon_0 \left(\chi^{(1)}\mathbf{E} + \chi^{(2)}\mathbf{E}^2 + \chi^{(3)}\mathbf{E}^3 + \cdots\right).$$

38.2 Types of SPDC

- **Type 0**: Signal and idler have the same polarization as the pump
- **Type I**: Signal and idler have the same polarization, orthogonal to pump
- **Type II**: Signal and idler have orthogonal polarizations

38.3 Types of Entanglement

Inherently entangled: energy–time, position–momentum, orbital angular momentum.

Forced entangled: polarization, time-bin.

38.4 Recipe for Entangled Single-Photon Sources

1. Creation of two orthogonal photon-pair states
2. Superposition
3. Splitting of photons

38.5 SPDC Phase-Matching Conditions

Energy conservation: $\omega_p = \omega_s + \omega_i$. Momentum conservation: $\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i$. Perfect phase matching: $\Delta k = k_p - k_s - k_i = 0$.

Critical phase matching (e.g. BBO): $\frac{n(\lambda_p,\theta)}{\lambda_p} = \frac{n(\lambda_s)}{\lambda_s} + \frac{n(\lambda_i)}{\lambda_i}$.

Quasi-phase matching (e.g. ppKTP, ppLN): $k_p = k_s + k_i + \frac{2\pi}{\Lambda}$.

38.6 Superposition and Bell States

The indistinguishability of orthogonal pair states allows superposition:

$$|\psi\rangle = |HH\rangle + e^{i\phi}\,|VV\rangle\,.$$

The four Bell states are

$$|\phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|HH\rangle \pm |VV\rangle), \qquad |\psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|HV\rangle \pm |VH\rangle).$$

38.7 Projection Probabilities

For polarization angles θ_1 and θ_2:

$$p(\theta_1,\theta_2) = \frac{1}{2}\cos^2(\theta_1 \mp \theta_2) \quad (\text{for } |\phi^{\pm}\rangle),$$
$$p(\theta_1,\theta_2) = \frac{1}{2}\sin^2(\theta_1 \pm \theta_2) \quad (\text{for } |\psi^{\pm}\rangle).$$

38.8 Polarization Visibility

Horizontal/Vertical basis:

$$V_{H/V} = \frac{N_{HH} + N_{VV} - N_{HV} - N_{VH}}{N_{HH} + N_{VV} + N_{HV} + N_{VH}} \times 100\%.$$

Diagonal/Antidiagonal basis:

$$V_{D/A} = \frac{N_{DD} + N_{AA} - N_{DA} - N_{AD}}{N_{DD} + N_{AA} + N_{DA} + N_{AD}} \times 100\%.$$

38.9 Sagnac Interferometer (Single-Crystal, Double-Pass)

For the collinear case:

$$|\psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|HV\rangle \pm |VH\rangle), \qquad |\phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|HH\rangle \pm |VV\rangle).$$

39 Polarization Entanglement from $\chi^{(2)}$

The polarization structure of the generated pairs is entirely determined by the crystal type and phase-matching geometry.

The SPDC Hamiltonian including polarization is

$$H_{\text{SPDC}} = i\hbar\kappa\left(e^{i\phi}\hat{a}_s^\dagger\hat{a}_i^\dagger - e^{-i\phi}\hat{a}_s\hat{a}_i\right),$$

where s, i label signal/idler including polarization. From the nonlinear polarization $P_i^{(2)} = \varepsilon_0\chi_{ijk}^{(2)}E_jE_k$, quantizing the field gives

$$H_{\text{SPDC}} \propto \sum_{s,i}\Gamma(s,i)\,\hat{a}_s^\dagger\hat{a}_i^\dagger + \text{h.c.}$$

39.1 Crystal-Type Dependence

Type 0: $H \to H + H$, so $H_{\text{SPDC}} \propto \hat{a}_{s,H}^\dagger\hat{a}_{i,H}^\dagger + \text{h.c.}$ (needs two crystals).

Type I: $H \to V + V$; both $\Gamma_{HH} \neq 0$ and $\Gamma_{VV} \neq 0$, $H_{\text{SPDC}} \propto \hat{a}_{s,H}^\dagger\hat{a}_{i,H}^\dagger + \hat{a}_{s,V}^\dagger\hat{a}_{i,V}^\dagger + \text{h.c.}$ (two crystals needed).

Type II: $\Gamma_{HV} \neq 0$, $\Gamma_{VH} \neq 0$; $H_{\text{SPDC}} \propto \hat{a}_{s,H}^\dagger\hat{a}_{i,V}^\dagger + \hat{a}_{s,V}^\dagger\hat{a}_{i,H}^\dagger + \text{h.c.}$ (inherent polarization entanglement).

39.2 State-Level Description

To first order in Γ, the two-photon state is

$$|\psi_{\text{pair}}\rangle \propto \sum_{s,i}\Gamma(s,i)\,|1_s, 1_i\rangle\,.$$

The photons are polarization-entangled if more than one $\Gamma(s,i)$ coefficient is nonzero.

Part V: SU(2) and SU(1,1) Algebras and Beam Splitters

Linear and parametric optical networks

40 Two-Mode Bosonic Realization of SU(2) and SU(1,1)

We consider two bosonic modes satisfying $[a_i, a_j^\dagger] = \delta_{ij}$, $[a_i, a_j] = 0$.

40.1 Total Number Operator

$$\hat{N} = a_1^\dagger a_1 + a_2^\dagger a_2.$$

40.2 SU(2) Generators (Schwinger Representation)

$$\begin{aligned}
\hat{J}_x &= \tfrac{1}{2}(a_1^\dagger a_2 + a_2^\dagger a_1), \\
\hat{J}_y &= \frac{1}{2i}(a_1^\dagger a_2 - a_2^\dagger a_1), \\
\hat{J}_z &= \tfrac{1}{2}(a_1^\dagger a_1 - a_2^\dagger a_2).
\end{aligned}$$

SU(2) algebra

$[\hat{J}_i, \hat{J}_j] = i\epsilon_{ijk}\hat{J}_k$

These satisfy $[\hat{J}_i, \hat{J}_j] = i\epsilon_{ijk}\hat{J}_k$. One can show $[\hat{N}, \hat{J}_x] = [\hat{N}, \hat{J}_y] = [\hat{N}, \hat{J}_z] = 0$, so the total photon number is a Casimir. The SU(2) Casimir is

$$\hat{\mathbf{J}}^2 = \frac{\hat{N}}{2}\left(\frac{\hat{N}}{2} + 1\right).$$

40.3 SU(1,1) Generators

$$\hat{K}_+ = a_1^\dagger a_2^\dagger, \quad \hat{K}_- = a_1 a_2, \quad \hat{K}_0 = \tfrac{1}{2}(a_1^\dagger a_1 + a_2^\dagger a_2 + 1).$$

SU(1,1) algebra

$[\hat{K}_+, \hat{K}_-] = -2\hat{K}_0$

Commutation relations:

$$[\hat{K}_0, \hat{K}_+] = +\hat{K}_+, \quad [\hat{K}_0, \hat{K}_-] = -\hat{K}_-, \quad [\hat{K}_+, \hat{K}_-] = -2\hat{K}_0.$$

Note the crucial minus sign in the last commutator (distinguishes SU(1,1) from SU(2)). The number difference $\hat{N}_- = a_1^\dagger a_1 - a_2^\dagger a_2$ commutes with $\hat{K}_0$ but not with $\hat{K}_\pm$. The SU(1,1) Casimir is

$$\hat{C} = \hat{K}_0^2 - \tfrac{1}{2}(\hat{K}_+\hat{K}_- + \hat{K}_-\hat{K}_+).$$

41 Quantum Light Interacting with a Beam Splitter

41.1 Classical Beam Splitter

Classically, $E_2 = rE_1$, $E_3 = tE_1$, with energy conservation $|r|^2 + |t|^2 = 1$.

41.2 Quantum Transformation

Assume a linear unitary transformation of annihilation operators:

$$a_i' = U_{ij} a_j.$$

Explicitly,

$$\begin{aligned} a_1' &= r a_1 + t a_2, \\ a_2' &= r' a_1 + t' a_2. \end{aligned}$$

41.3 Preservation of Commutation Relations

Requiring $[a_i', a_j'^\dagger] = \delta_{ij}$ gives $UU^\dagger = \mathbb{I}$, so $U \in \mathrm{U}(2)$. Photon number conservation further restricts this to $U \in \mathrm{SU}(2)$.

41.4 Parameterisation of U(2)

Any $U \in \mathrm{U}(2)$ can be written as $U = e^{i\phi} V$ with $V \in \mathrm{SU}(2)$; the global phase is physically irrelevant. A general SU(2) matrix:

$$U_{BS} = \begin{pmatrix} \cos\theta\, e^{i\phi_t} & \sin\theta\, e^{i\phi_r} \\ -\sin\theta\, e^{-i\phi_r} & \cos\theta\, e^{-i\phi_t} \end{pmatrix}.$$

41.5 Generator Form

Since $U \in \mathrm{SU}(2)$, we can write $U = e^{-iG}$ with $G = \boldsymbol{\theta} \cdot \boldsymbol{J}$ using the Schwinger generators. Thus the beam splitter Hamiltonian must be quadratic in ladder operators:

$$H \propto a_1^\dagger a_2 + a_2^\dagger a_1.$$

41.6 Heisenberg Evolution

For infinitesimal $U = 1 - i\epsilon G$,

$$a' = a - i\epsilon [G, a].$$

41.7 Most General Number-Conserving Hamiltonian

A quadratic Hamiltonian $\hat{H} = \sum_{m,n} h_{mn} a_m^\dagger a_n$ gives

$$[\hat{H}, a_k] = -\sum_n h_{kn} a_n,$$

so $h = G$.

42 Beam Splitter as an SU(2) Transformation

Choosing $G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

$$\hat{H}_{BS} = \kappa(a_1^\dagger a_2 + a_2^\dagger a_1) = 2\kappa \hat{J}_x, \quad U_{BS} = e^{-i\theta \hat{J}_x}.$$

Beam splitter

$\hat{H}_{BS} = \hbar\kappa \hat{J}_x$

The Heisenberg equations of motion are $\dot{a}_1 = -i\kappa a_2$, $\dot{a}_2 = -i\kappa a_1$, with solution

$$\begin{pmatrix} a_1(t) \\ a_2(t) \end{pmatrix} = \begin{pmatrix} \cos(\kappa t) & -i\sin(\kappa t) \\ -i\sin(\kappa t) & \cos(\kappa t) \end{pmatrix} \begin{pmatrix} a_1(0) \\ a_2(0) \end{pmatrix}.$$

42.1 Phase Shifter

$\hat{J}_z = \frac{1}{2}(a_1^\dagger a_1 - a_2^\dagger a_2)$ generates relative phase shifts:

$$a_1' = e^{-i\phi/2} a_1, \qquad a_2' = e^{+i\phi/2} a_2.$$

Summary: $\hat{J}_z \to$ phase shifter; $\hat{J}_x, \hat{J}_y \to$ beam splitter; $H_{sq} = \hbar\kappa\hat{K}_x \to$ SU(1, 1).

42.2 General SU(2) Form and Decomposition

A general SU(2) matrix $U = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}$ with $|\alpha|^2 + |\beta|^2 = 1$ decomposes as

$$U_{BS} = D_L\, R(\theta)\, D_R,$$

where D_L, D_R are phase matrices and $R(\theta)$ is a rotation: thus a beam splitter equals *(phase shift)×(mode mixing)×(phase shift)*.

For small θ, $R(\theta) \approx e^{-i\theta\sigma_y}$.

43 Application to Quantum States

43.1 Vacuum

Beam splitter leaves vacuum invariant: $|0\rangle_0 |0\rangle_1 \to |0\rangle_2 |0\rangle_3$.

43.2 Single-Photon Input

Input: $|\psi_{\text{in}}\rangle = |0\rangle_0 |1\rangle_1 = a_1^\dagger |0\rangle$.

Beam splitter transformations:

$$a_0^\dagger \to \tfrac{1}{\sqrt{2}}(a_2^\dagger + ia_3^\dagger), \quad a_1^\dagger \to \tfrac{1}{\sqrt{2}}(ia_2^\dagger + a_3^\dagger).$$

Output:

$$|\psi_{\text{out}}\rangle = \frac{1}{\sqrt{2}}(i\,|1\rangle_2 |0\rangle_3 + |0\rangle_2 |1\rangle_3)\,.$$

This is a **path-entangled** single-photon state.

43.3 Coherent State Input

Input: $|0\rangle_0 |\alpha\rangle_1$. The output is

$$|\psi_{\text{out}}\rangle = \left|\tfrac{i\alpha}{\sqrt{2}}\right\rangle_2 \otimes \left|\tfrac{\alpha}{\sqrt{2}}\right\rangle_3.$$

Coherent states remain **separable** under a beam splitter.

44 Single-Photon Interferometer (Balanced Mach–Zehnder)

In a balanced MZI both beam splitters are 50:50. A phase ϕ in one arm shifts probability between the two output ports. Interference is of *probability amplitudes*, not intensities.

State after BS1 (photon enters lower port, mode 1):

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(i\,|1_0, 0_1\rangle + |0_0, 1_1\rangle)\,.$$

After relative phase shift $e^{\pm i\phi/2}$:

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}\left(ie^{i\phi/2}\,|1_0, 0_1\rangle + e^{-i\phi/2}\,|0_0, 1_1\rangle\right).$$

After BS2:

$$|\psi_{\text{out}}\rangle = i\cos\left(\tfrac{\phi}{2}\right)|1_2, 0_3\rangle - i\sin\left(\tfrac{\phi}{2}\right)|0_2, 1_3\rangle\,.$$

Detection probabilities:

$$P(D_2) = \cos^2\left(\tfrac{\phi}{2}\right), \quad P(D_3) = \sin^2\left(\tfrac{\phi}{2}\right).$$

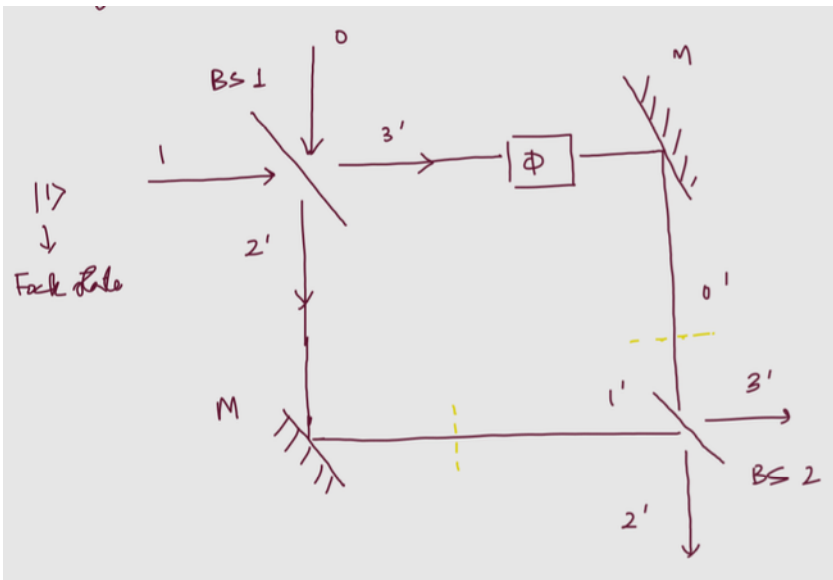

Figure 3: Single-photon Mach–Zehnder interferometer. BS1 and BS2 are 50:50 beam splitters; M are mirrors; ϕ is the relative phase between the two arms.

44.1 Single-Photon Interference

Classical particle case: no interference fringes. Classical wave case: interference fringes appear. Quantum case: single detector clicks at any given time; accumulated counts show interference fringes. **Key idea:** interference of probability amplitudes.

Note

The photon's temporal coherence length must exceed the path-length difference for interference to occur.

45 Hong–Ou–Mandel Effect

Input state: $|1\rangle_0 |1\rangle_1 = a_0^\dagger a_1^\dagger |0\rangle$.

Beam splitter transformation:

$$a_0^\dagger \to \tfrac{1}{\sqrt{2}}(a_2^\dagger + i a_3^\dagger), \quad a_1^\dagger \to \tfrac{1}{\sqrt{2}}(i a_2^\dagger + a_3^\dagger).$$

Output state:

$$\frac{1}{2}(i a_2^{\dagger 2} + i a_3^{\dagger 2}) |0\rangle = \frac{i}{\sqrt{2}}(|2\rangle_2 |0\rangle_3 + |0\rangle_2 |2\rangle_3).$$

Physical insight

The HOM dip is the standard benchmark for photon indistinguishability. Its visibility directly quantifies the purity and mode overlap of the two photons.

Hong–Ou–Mandel Effect

Coincidence probability $= 0$. Two indistinguishable photons always exit the same port (bunching). This is a purely quantum interference effect.

46 N-Photon State at a Beam Splitter

Input Fock state: $|\psi_{\text{in}}\rangle = \frac{(a_1^\dagger)^N}{\sqrt{N!}} |0\rangle$.

After beam splitter:

$$|\psi_{\text{out}}\rangle = \sum_{k=0}^{N} \sqrt{\binom{N}{k}} \frac{i^k}{2^{N/2}} |k\rangle_2 |N-k\rangle_3 .$$

Expectation values:

$$\langle k \rangle = \frac{N}{2}, \quad \langle k^2 \rangle = \frac{N^2}{4} + \frac{N}{4}, \quad \Delta k^2 = \frac{N}{4}.$$

Relative fluctuations: $\frac{\sqrt{\Delta k^2}}{\langle k \rangle} = \frac{1}{\sqrt{N}}$.

Large-N limit (Central Limit Theorem):

$$P(k) \approx \frac{1}{\sqrt{2\pi(N/4)}} \exp\left(-\frac{(k - N/2)^2}{2(N/4)}\right).$$

47 Standard Quantum Limit for Phase Measurement

The Mach–Zehnder transformation in matrix form is

$$\frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} 0 & -ie^{-i\phi/2} \\ e^{i\phi/2} & 0 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \begin{pmatrix} \cos(\phi/2) & \sin(\phi/2) \\ -\sin(\phi/2) & \cos(\phi/2) \end{pmatrix}.$$

SQL

$\Delta\phi = 1/\sqrt{N}$

Define the observable $\hat{D} = a_3^\dagger a_3 - a_2^\dagger a_2$. After the MZI transformation,

$$\hat{D} = \cos\phi(a_1^\dagger a_1 - a_0^\dagger a_0) + \sin\phi(a_0^\dagger a_1 + a_1^\dagger a_0).$$

47.1 Case I: Coherent Input + Vacuum

$$\langle \hat{D} \rangle = N \cos\phi, \quad \left| \frac{d\langle \hat{D} \rangle}{d\phi} \right| = N \sin\phi.$$

Maximum sensitivity at $\phi = \pi/2$. Near $\phi = \pi/2$, writing $a_1 = \alpha + \delta\hat{a}$,

$$\hat{D} \approx \alpha(\hat{a}_0^\dagger + \hat{a}_0) = 2\sqrt{N}\,\hat{X}_0, \quad \Delta D^2 = 4N\,\Delta X_0^2.$$

Phase uncertainty via error propagation:

$$\Delta\phi = \frac{\Delta D}{|d\langle \hat{D} \rangle / d\phi|} = \frac{2\sqrt{N}\sqrt{1/4}}{N} = \frac{1}{\sqrt{N}}.$$

The vacuum entering the unused port sets the noise floor. This is why injecting squeezed vacuum there reduces the phase uncertainty.

Standard Quantum Limit (SQL)

$\Delta\phi = 1/\sqrt{N}$ is the shot-noise limit for N photons with vacuum in the second port.

47.2 Case II: Coherent Input + Squeezed Vacuum

If squeezing is in the X quadrature: $\Delta X_0^2 = e^{-2r}/4$, then

$$\Delta\phi \approx \frac{e^{-r}}{\sqrt{N}} \quad \text{for } N \gg \sinh^2 r.$$

Thus squeezing improves phase sensitivity beyond the SQL.

48 Optical Attenuation (Beam Splitter Model)

Any optical loss channel is modeled as a beam splitter coupling the signal mode to a vacuum reservoir. With transmission η:

$$a_2 = \sqrt{\eta}\,a_0 + i\sqrt{1-\eta}\,a_1, \qquad a_3 = i\sqrt{1-\eta}\,a_0 + \sqrt{\eta}\,a_1.$$

Any optical loss channel — absorption, scattering, imperfect detectors — can be modeled as a beam splitter coupling the signal to a vacuum reservoir.

48.1 First Moment

For vacuum in port 1: $\langle a_2 \rangle = \sqrt{\eta}\,\langle a_0 \rangle$. A coherent input attenuates as $|\alpha\rangle \to |\sqrt{\eta}\,\alpha\rangle$.

48.2 Quadrature Variance

$$\Delta(X_\theta^{(2)})^2 = \eta\,\Delta(X_\theta^{(0)})^2 + (1-\eta)\,\frac{1}{4}.$$

Optical Attenuation Adds Vacuum Noise

Every photon lost is replaced by a vacuum-noise photon. Attenuation introduces a mixture with the vacuum, degrading any non-classical feature.

48.3 Squeezed Vacuum Under Attenuation

For a squeezed state $\Delta X^2 = e^{-2r}/4$:

$$\Delta(X^{(2)})^2 = \eta \frac{e^{-2r}}{4} + (1-\eta)\frac{1}{4}.$$

In the limit $\eta \to 0$: $\Delta(X^{(2)})^2 = 1/4$ — all squeezing is destroyed.

48.4 Photon Number Statistics Under Attenuation

$$\begin{aligned} \langle \hat{n}_2 \rangle &= \eta \langle \hat{n}_0 \rangle, \\ \langle \Delta n_2^2 \rangle &= \eta^2 \langle \Delta n_0^2 \rangle + \eta(1-\eta)\langle \hat{n}_0 \rangle. \end{aligned}$$

For strong attenuation $\eta \ll 1$: $\langle \Delta n_2^2 \rangle \approx \langle n_2 \rangle$ — the **Poissonian limit**.

Part VI: Balanced Homodyne Detection

Measuring quadratures and quantum noise

49 Balanced Homodyne Detection

49.1 Classical Description

Balanced homodyne detection cancels LO noise in the difference current, leaving only the quadrature noise of the signal amplified by $|\beta|$.

Consider an unknown signal field $E_s = E_s \cos(\omega t + \phi_s)$ and a strong local oscillator (LO) $E_{\text{LO}} = E_{\text{LO}} \cos(\omega t + \phi_{\text{LO}})$ combined on a 50:50 beam splitter. The difference current is

$$\langle I_- \rangle = 2E_s E_{\text{LO}} \sin(\phi_s - \phi_{\text{LO}}).$$

Key observations:

- Linear amplification of the signal proportional to $|E_{\text{LO}}|$.
- $\phi_{\text{LO}} = 0 \Rightarrow$ sine quadrature; $\phi_{\text{LO}} = \pi/2 \Rightarrow$ cosine quadrature.
- LO noise cancels in the difference signal.

49.2 Quantum Description

The positive-frequency field operator is $\hat{E}^{(+)} = \epsilon\, \hat{a}\, e^{-i\omega t}$. After the beam splitter,

$$\hat{I}_- = \hat{I}_2 - \hat{I}_3 = i|\epsilon|^2(\hat{a}_0^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_0).$$

Assuming the LO is in coherent state $|\beta\rangle$, $\beta = |\beta|e^{i\theta}$:

$$\langle I_- \rangle = 2K|\beta|\langle \hat{P}_\theta \rangle,$$

where $\hat{P}_\theta$ is the quadrature operator at angle θ.

49.3 Noise

The variance of the difference current is

$$\Delta I_-^2 = 4K|\beta|^2 \Delta(\hat{P}_\theta)^2 + K^2 \langle a_0^\dagger a_0 \rangle.$$

The first term is amplified quadrature noise; the second is residual noise from commutation relations.

49.4 Shot Noise vs. Squeezing

With vacuum in port 0: $\langle \Delta I_-^2 \rangle = K^2|\beta|^2$.

With squeezed vacuum in port 0:

$$\langle \Delta I_-^2 \rangle = K^2|\beta|^2 e^{-2r} + K^2 \sinh^2 r.$$

For strong LO the second term is negligible.

49.5 Squeezing in dB

$$\text{squeezing (dB)} = -10\log_{10}\left(\frac{\langle \Delta I_-^2 \rangle}{\langle \Delta I_-^2 \rangle_{\text{SN}}}\right) = -10\log_{10}(e^{-2r}) \approx 8.7r.$$

Squeezing in dB

$\approx -10\log_{10}(e^{-2r}) \approx 8.7r$

50 Balanced Homodyne Setup for Quadrature Measurement

50.1 Realistic Homodyne

Suppose a source produces 10 dB squeezing but only ~ 6 dB is measured. Possible reasons: noise, optical attenuation, detector inefficiency, mode mismatch.

50.2 Classical Description of Mode Mismatch

Signal field $E_s(\mathbf{r}) = E_0\, g(\mathbf{r})$; LO $E_{\text{LO}}(\mathbf{r}) = E_L\, f(\mathbf{r})$. The difference current is $I_- \propto \int f^*(\mathbf{r})\, g(\mathbf{r})\, d^2r$.

The modal overlap: $\eta = \langle f|g\rangle$ ($\eta = 1$: perfect; $\eta = 0$: no interference).

50.3 Quantum Upgrade

Define multimode operators

$$\hat{A}_f = \int f(\omega)\, \hat{a}(\omega)\, d\omega, \quad \hat{A}_g = \int g(\omega)\, \hat{a}(\omega)\, d\omega.$$

Decompose the LO mode:

$$\hat{A}_f = \eta \hat{A}_g + \sqrt{1-\eta^2}\, \hat{A}_\perp.$$

The measured quadrature becomes

$$X_{\text{meas}} = \eta X_{\text{signal}} + \sqrt{1-\eta^2}\, X_{\text{vac}},$$

with variance

$$\langle (\Delta X_{\text{meas}})^2 \rangle = \eta^2 \Delta X_{\text{signal}}^2 + (1-\eta^2)\frac{1}{4}.$$

51 Multimode Quantum Optics

In reality the EM field contains infinitely many modes:

$$E^{(+)}(\mathbf{r}, t) = \sum_k \epsilon_k \hat{a}_k u_k(\mathbf{r}) e^{-i\omega_k t}.$$

A *mode* is not a photon; a photon is an excitation of a mode: $|1_k\rangle = \hat{a}_k^\dagger |0\rangle$.

A single-mode description is valid only when a cavity, filter, or detector selects a single spatial–temporal mode. In practice, broadband sources require a multimode treatment.

51.1 When is the Single-Mode Approximation Valid?

- Optical cavity selects a single mode
- Optical filtering
- Perfect modal overlap between field and detector
- Narrow-band detection

51.2 Mode Functions

Mode functions $u_k(\mathbf{r}, t)$ satisfy normalization, orthogonality, and completeness:

$$\int d^2r\, dt\, |u_j|^2 = 1, \quad \iint u_j^* u_k = \delta_{jk}, \quad \sum_k u_k^*(\mathbf{r}, t) u_k(\mathbf{r}', t') = \delta^{(2)}(\mathbf{r} - \mathbf{r}')\delta(t - t').$$

51.3 Multimode Field Operator

$$\hat{E}^{(+)}(\mathbf{r}, t) = \sum_k \hat{a}_k u_k(\mathbf{r}, t), \quad [\hat{a}_j, \hat{a}_k^\dagger] = \delta_{jk}.$$

Equal-time commutation:

$$[\hat{E}^{(+)}(\mathbf{r}, t), \hat{E}^{(-)}(\mathbf{r}', t)] = \delta^{(2)}(\mathbf{r} - \mathbf{r}').$$

51.4 Wave-Packet Mode

Taking normalized $f(\omega)$, define

$$\hat{A}_f = \int f(\omega)\hat{a}(\omega)e^{-i\omega t}\, d\omega, \quad [\hat{A}_f, \hat{A}_f^\dagger] = \int d\omega\, |f(\omega)|^2 = 1.$$

A wave packet is a new basis choice for modes. A non-monochromatic single photon is $|\psi\rangle = \hat{A}_f^\dagger |0\rangle = \sum_k c_k |1_k\rangle$.

51.5 Annihilation Operator for an Arbitrary Mode

$$\hat{a}_f = \int d^2r \int dt\, f^*(\mathbf{r}, t)\, \hat{E}^{(+)}(\mathbf{r}, t) = \sum_k c_k \hat{a}_k, \quad c_k = \langle f | u_k \rangle.$$

If $\langle f | g \rangle = 0$: $[\hat{a}_f, \hat{a}_g^\dagger] = 0$ (orthogonal modes commute).

51.6 Vacuum State and Shot Noise

$\hat{a}_n |\text{vac}\rangle = 0$ for all n, so $\hat{a}_f |\text{vac}\rangle = 0$. Vacuum variance is basis-independent:

$$\langle \text{vac} | (\Delta \hat{X}_\theta)^2 | \text{vac} \rangle = \frac{1}{4}.$$

Shot noise is invariant to basis change.

52 Multimode Homodyne Detection

52.1 Step 1: Mode Decomposition

$$\hat{E}_0^{(+)}(\mathbf{r}, t) = \hat{a}_0 u_s(\mathbf{r}, t) + \sum_{k=1}^{\infty} \hat{v}_k w_k(\mathbf{r}, t).$$

Classical LO: $E_1^{(+)} = \alpha_{\text{LO}} u_{\text{LO}}(\mathbf{r}, t)$.
Overlaps: $\xi = \int u_s^* u_{\text{LO}}, \quad \xi_k = \int u_{\text{LO}}^* w_k$.

52.2 Step 2: Beam Splitter

$$E_2^{(+)} = \tfrac{1}{\sqrt{2}}[\hat{a}_0 u_s + \textstyle\sum_k \hat{v}_k w_k + i\alpha_{\rm LO} u_{\rm LO}],$$
$$E_3^{(+)} = \tfrac{1}{\sqrt{2}}[i\hat{a}_0 u_s + i\textstyle\sum_k \hat{v}_k w_k + \alpha_{\rm LO} u_{\rm LO}].$$

52.3 Step 3: Intensity and Difference Current

Computing $\hat{n}_j = \int d^2r \int dt\, \hat{E}_j^{(-)} \hat{E}_j^{(+)}$ and forming $\hat{I} = \hat{n}_2 - \hat{n}_3$:

$$\hat{I} = i\alpha_{\rm LO}\xi^* \hat{a}_0^\dagger - i\alpha_{\rm LO}^* \xi \hat{a}_0 + i\alpha_{\rm LO} \sum_k \hat{v}_k^\dagger \xi_k^* - i\alpha_{\rm LO}^* \sum_k \hat{v}_k \xi_k.$$

52.4 LO Mode Projected Operator

$$\hat{A}_{\rm LO} = \int d^2r \int dt\, u_{\rm LO}^* \hat{E}_0^{(+)} = \hat{a}_0 \xi + \sum_k \hat{v}_k \xi_k.$$

53 Mode Mismatch in Homodyne

The difference current reduces to

$$\hat{I}^- = i\alpha_{\rm LO}\hat{A}_{\rm LO}^\dagger - i\alpha_{\rm LO}^* \hat{A}_{\rm LO} = 2|\alpha_{\rm LO}|\hat{P}_\theta^{(\rm proj)},$$

where

$$\hat{P}_\theta^{(\rm proj)} = \frac{\hat{A}_{\rm LO} e^{-i\theta} - \hat{A}_{\rm LO}^\dagger e^{i\theta}}{2i}.$$

Homodyne captures only that part of the signal aligned with the LO mode.

53.1 Mode Decomposition

$$\eta_{mm} = |\xi_g|^2 = \left| \int d^2r \int dt\, u_{\rm LO}^* u_s \right|^2.$$

Defining a normalized vacuum operator $\hat{v}_\perp = (1-\eta_{mm})^{-1/2} \sum_k \hat{v}_k \xi_k$,

$$\hat{A}_{\rm LO} = \sqrt{\eta_{mm}} e^{i\phi_s} \hat{a}_0 + \sqrt{1-\eta_{mm}}\, \hat{v}_\perp.$$

The measured quadrature is

$$\hat{P}_\theta^{(\rm proj)} = \sqrt{\eta_{mm}} \hat{P}_{\theta-\phi_s}^{(s)} + \sqrt{1-\eta_{mm}}\, \hat{P}_\theta^\perp.$$

53.2 Variance and Shot Noise

Shot noise: $\Delta I_-^2 = |\alpha_{\rm LO}|^2$.
Squeezed input with mode mismatch:

$$\Delta I_-^2 = |\alpha_{\rm LO}|^2 \eta_{mm} e^{-2r} + |\alpha_{\rm LO}|^2 (1-\eta_{mm}).$$

53.3 Meaning of η_{mm}

The fringe visibility for two equal-amplitude coherent fields is

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = |\xi_g| = \sqrt{\eta_{mm}}.$$

53.4 LO Phase Noise

If the LO phase fluctuates: $\theta \to \theta + \delta\theta(t)$,

$$\mathrm{Var}(\hat{P}_{\theta+\delta\theta}) = \Delta P_\theta^2 \cos^2 \delta\theta + \Delta X_\theta^2 \sin^2 \delta\theta.$$

For Gaussian phase noise $P(\delta\theta) = \frac{1}{\sqrt{2\pi\sigma_\theta^2}} e^{-\delta\theta^2/2\sigma_\theta^2}$,

$$\Delta P_{\theta,\mathrm{meas}}^2 = \Delta P_\theta^2 (1 - \sigma_\theta^2) + \Delta X_\theta^2 \sigma_\theta^2.$$

Part VII: Quantum Key Distribution

Securing communication with quantum mechanics

54 Classical Cryptography (One-Time Pad)

Binary encoding via XOR:

$$0 \oplus 0 = 0, \quad 0 \oplus 1 = 1, \quad 1 \oplus 0 = 1, \quad 1 \oplus 1 = 0.$$

The one-time pad is information-theoretically secure (Shannon, 1949) but requires a key as long as the message and a perfectly secure key distribution channel.

Issues: (i) need secure key transmission; (ii) key can be guessed via brute force.

55 Quantum Key Distribution

Quantum mechanics helps via:

- Measurement disturbs the system (no-cloning theorem)
- Eavesdropping is detectable

QKD provides reliable key distribution and can detect eavesdroppers, though it does not provide unconditional security against all attacks without assumptions.

56 BB84 Protocol

Basis encoding:

Bit	$+$ basis	$\times$ basis
0	$\lvert\downarrow\rangle$	$\lvert\rightarrow\rangle$
1	$\lvert\uparrow\rangle$	$\lvert\nearrow\rangle$

Alice randomly chooses basis and sends bits. Bob randomly chooses basis for measurement. After measurement, they compare bases publicly and keep only matched-basis bits (*sifting*).

56.1 Eavesdropping Detection

If Eve intercepts:

$$P(\text{error}) = 50\% \times 50\% = 25\%.$$

Eve must guess the basis, which she gets right only 50% of the time. When wrong, she disturbs the state. The resulting QBER of 25% is a detection threshold.

Thus an error rate $\geq 25\%$ indicates eavesdropping.

57 Two-Photon Case (Improved Protocol)

Alice sends 2 photons per bit.

57.1 Case Analysis

If Eve chooses the correct basis: both photons go to the same detector — no error.

If Eve chooses the wrong basis: outcomes are probabilistic, giving 4 possible detection combinations.

Fraction of altered photons: $P(\text{altered}) = \frac{1}{2} \times \frac{2}{4} = \frac{1}{4}$.

Fraction of no photon sent: $= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, so 75% contain a photon.

57.2 Conditional Probability and Error Rate

$$P(\text{altered} \mid \text{photon sent}) = \frac{1/4}{3/4} = \frac{1}{3}.$$

With Bob choosing the correct basis and $1/3$ of photons altered:

$$P_{\text{error}} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6} \approx 16.7\%.$$

Conclusion: error rate reduced from 25% to 16.7%.

57.3 Optical Setup

50:50 beam splitter + detectors, with input beam, half-wave plate (HWP), mirror, and detectors D_1, D_2, D_3, D_4.

Part VIII: Atom–Light Interaction

From Fermi's golden rule to the Jaynes–Cummings model

58 Emission and Absorption of Light by Atoms

The total Hamiltonian is

$$\hat{H} = \hat{H}_{\text{atom}} + \hat{H}_F + \hat{H}_{\text{int}}.$$

The interaction follows from minimal coupling:

$$\hat{H}_{\text{int}} = -\mathbf{d} \cdot \mathbf{E} = -e(\mathbf{r} - \mathbf{r}_0) \cdot \mathbf{E},$$

derived via $\frac{(\mathbf{p}-e\mathbf{A})^2}{2m} = \frac{\mathbf{p}^2}{2m} - \frac{e}{m}\mathbf{A} \cdot \mathbf{p} + \frac{e^2 A^2}{2m}$.

59 Lagrangian of Electron + EM Field

$$L = \frac{1}{2}m\dot{\mathbf{r}}^2 + e\phi(\mathbf{r}, t) - e\dot{\mathbf{r}} \cdot \mathbf{A}(\mathbf{r}, t) + \frac{\epsilon_0}{2}\int d^3r \left[|\dot{\mathbf{A}} + \nabla\phi|^2 - c^2|\nabla \times \mathbf{A}|^2\right].$$

The Lagrangian is the starting point for deriving the canonical momentum and then the Hamiltonian via a Legendre transform.

The canonical momentum is $\mathbf{p} = m\dot{\mathbf{r}} - e\mathbf{A}$, and the Hamiltonian is

$$H = \frac{(\mathbf{p} + e\mathbf{A})^2}{2m} - e\phi.$$

60 Coulomb Gauge

In Coulomb gauge: $\nabla \cdot \mathbf{A} = 0$, $\nabla^2\phi = \rho/\epsilon_0$. Under gauge transformation $\mathbf{A}' = \mathbf{A} + \nabla\chi$, $\phi' = \phi - \partial_t\chi$, the condition $\nabla \cdot \mathbf{A}' = 0$ requires $\nabla^2\chi = -\nabla \cdot \mathbf{A}$.

The full Hamiltonian is

$$H_{\text{tot}} = \frac{(\mathbf{p} + e\mathbf{A})^2}{2m} + V(\mathbf{r}) + H_F, \quad H_F = \frac{\epsilon_0}{2}\int d^3r\,(E^2 + c^2B^2).$$

Note

The choice $\nabla \cdot \hat{\mathbf{A}} = 0$ (Coulomb gauge) is a *gauge condition*, not a physical constraint. Physical observables are gauge-invariant.

60.1 Quantization and Mode Expansion

Canonical variable for the field is $\mathbf{A}$ subject to $\nabla \cdot \mathbf{A} = 0$. Conjugate momentum $\Pi_i = \epsilon_0 E_i(\mathbf{r})$, with commutation $[\hat{A}_i(\mathbf{r}), \hat{\Pi}_j(\mathbf{r}')] = i\hbar\delta_{ij}\delta(\mathbf{r} - \mathbf{r}')$.

Mode expansion:

$$\hat{\mathbf{A}}(\mathbf{r}) = \sum_{\mathbf{k},\lambda} \sqrt{\frac{\hbar}{2\omega_k \epsilon_0 V}}\, \boldsymbol{\epsilon}_{\mathbf{k}\lambda} \left(\hat{a}_{\mathbf{k}\lambda} e^{i\mathbf{k}\cdot\mathbf{r}} + \hat{a}^\dagger_{\mathbf{k}\lambda} e^{-i\mathbf{k}\cdot\mathbf{r}} \right),$$

$$\hat{\mathbf{\Pi}}(\mathbf{r}) = -i \sum_{\mathbf{k},\lambda} \sqrt{\frac{\epsilon_0 \hbar \omega_k}{2V}}\, \boldsymbol{\epsilon}_{\mathbf{k}\lambda} \left(\hat{a}_{\mathbf{k}\lambda} e^{i\mathbf{k}\cdot\mathbf{r}} - \hat{a}^\dagger_{\mathbf{k}\lambda} e^{-i\mathbf{k}\cdot\mathbf{r}} \right).$$

The field Hamiltonian is

$$H_F = \sum_{\mathbf{k},\lambda} \hbar\omega_k \left(a^\dagger_{\mathbf{k}\lambda} a_{\mathbf{k}\lambda} + \frac{1}{2} \right).$$

61 Interaction Hamiltonian

Expanding $(\mathbf{p} + e\mathbf{A})^2$ and using $\nabla \cdot \mathbf{A} = 0$ (so $\mathbf{p} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{p}$) and neglecting A^2:

$$H = \frac{p^2}{2m} + \frac{e}{m} \mathbf{p} \cdot \mathbf{A} + V(\mathbf{r}) + H_F.$$

62 Göppert-Mayer Transformation

We transform from $\mathbf{p} \cdot \mathbf{A}$ to $\mathbf{d} \cdot \mathbf{E}$ via the unitary operator

$$\hat{R} = \exp\left[\frac{ie}{\hbar} (\hat{\mathbf{r}} - \mathbf{r}_0) \cdot \hat{\mathbf{A}}(\mathbf{r}_0) \right].$$

The Göppert-Mayer (Power–Zienau–Woolley) transformation is a unitary change of gauge that converts the $\mathbf{p} \cdot \mathbf{A}$ interaction to the more physical $\mathbf{d} \cdot \mathbf{E}$ form.

Under this transformation $|\psi'\rangle = \hat{R}|\psi\rangle$, the new Hamiltonian is $H' = RHR^\dagger - i\hbar R \partial_t R^\dagger$.

62.1 Operator Transformations

$$R\hat{\mathbf{p}}R^\dagger = \hat{\mathbf{p}} - e\mathbf{A}(\mathbf{r}_0), \quad RV(\mathbf{r})R^\dagger = V(\mathbf{r}).$$

Using the commutator $[A_i(\mathbf{r}_0), H_F] = -i\hbar E_i(\mathbf{r}_0)$:

$$[\Lambda, H_F] = \frac{e}{\hbar} \sum_i (r_i - r_{0i}) [A_i(\mathbf{r}_0), H_F] = i\, \mathbf{d} \cdot \mathbf{E}.$$

In the dipole approximation $\mathbf{A}(\mathbf{r}) \approx \mathbf{A}(\mathbf{r}_0)$:

$$R\left(H_{\text{atom}} + \frac{e}{m} \mathbf{A} \cdot \mathbf{p} \right) R^\dagger = \frac{p^2}{2m} + V(\mathbf{r}).$$

62.2 Final Hamiltonian

$$\boxed{H' = H_{\text{atom}} + H_F - \mathbf{d} \cdot \mathbf{E}(\mathbf{r}_0).}$$

63 Interaction with Classical Light: Perturbation Theory

With $E(t) = E_0 \cos(\omega t)$ for $t > 0$ and initial state $|\psi(0)\rangle = |i\rangle$, expand $|\psi(t)\rangle = \sum_k C_k(t) e^{-iE_k t/\hbar} |k\rangle$.

Time-dependent perturbation theory is valid when the field is weak: $\Omega_R \ll \omega_0$. Near resonance with strong driving, a non-perturbative approach is needed.

63.1 Perturbation Expansion

$$C_i^{(r)}(t) = \frac{1}{i\hbar}\sum_k C_k^{(r-1)} e^{-i\omega_{ki}t}(H_I)_{ik}.$$

First order: $C_l^{(0)}(t) = \delta_{il}$, so

$$C_l^{(1)}(t) = \frac{1}{i\hbar} e^{i\omega_{li}t}(H_I)_{li}.$$

63.2 Selection Rule

$\langle l|\hat{\mathbf{d}}|i\rangle \neq 0$ requires states of opposite parity. For $l = i$: $(H_I)_{ii} = 0$.

63.3 Transition to Final State f

$$C_f^{(1)}(t) = \frac{(\mathbf{d}\cdot\mathbf{E}_0)_{fi}}{2i\hbar}\left[\frac{e^{i(\omega_{fi}-\omega)t}-1}{i(\omega_{fi}-\omega)} + \frac{e^{i(\omega_{fi}+\omega)t}-1}{i(\omega_{fi}+\omega)}\right].$$

63.4 Rotating Wave Approximation

Quasi-resonant $\omega \approx \omega_{fi}$; drop anti-resonant term:

$$C_f^{(1)}(t) = -\frac{(\mathbf{d}\cdot\mathbf{E}_0)_{fi}}{2\hbar}\frac{e^{i(\omega_{fi}-\omega)t}-1}{\omega_{fi}-\omega}.$$

63.5 Transition Probability

$$P_{i\to f} = |C_f^{(1)}(t)|^2 = \frac{|(\mathbf{d}\cdot\mathbf{E}_0)_{fi}|^2}{4\hbar^2}\frac{\sin^2[(\omega_{fi}-\omega)t/2]}{[(\omega_{fi}-\omega)/2]^2}.$$

Defining $g_t(\Omega) = \frac{\sin^2(\Omega t/2)}{(\Omega/2)^2}$ (peak t^2 at $\Omega = 0$, width $\sim 2\pi/t$, area $\sim 2\pi t$), and $\delta_t(\Omega) = g_t(\Omega)/(2\pi t)$:

$$P_{i\to f}^{(1)} = \frac{|(\mathbf{d}\cdot\mathbf{E}_0)_{fi}|^2}{4\hbar^2}(2\pi t)\,\delta_t(\Omega).$$

63.6 Population of Levels

Only levels with $|E_f - E_i - \hbar\omega| < 2\pi\hbar/t$ are significantly populated.

63.7 Continuum Case and Fermi's Golden Rule

In the continuum,

$$P_{i\to f}^{(1)} = 2\pi t\, F(\omega_{fi}).$$

The transition rate is $\Gamma_{i\to f} = P_{i\to f}/t = 2\pi F(\omega_{fi})$.

Fermi's Golden Rule

$$\Gamma = \frac{2\pi}{\hbar} \left|\langle f|\mathbf{d}\cdot\mathbf{E}_0|i\rangle\right|^2 \rho(\omega_{fi}),$$

where $\rho(\omega)$ is the density of final states.

63.8 Probability of Staying in Initial State

For large t: $P_i(t) = 1 - \Gamma t$, confirming exponential decay.

63.9 Perturbation Results Summary

$$P^{(1)}_{i\to f} \propto \begin{cases} t^2 & \Delta = 0, \\ \text{oscillatory} & \Delta = \omega - \omega_0 \neq 0. \end{cases}$$

64 Non-Perturbative Solution: Two-Level Approximation

Near resonance excitation drives transitions strongly, requiring a non-perturbative treatment. Restrict to two levels $|a\rangle, |b\rangle$ with $E_b - E_a = \hbar\omega_0$:

The two-level approximation is justified when one transition is much closer to resonance than all others. The remaining transitions are far off-resonance and contribute negligibly.

$$H_{\text{atom}} = \begin{pmatrix} 0 & 0 \\ 0 & \hbar\omega_0 \end{pmatrix}.$$

The interaction gives $(H_I)_{ba} = \hbar\Omega_R \cos(\omega t + \phi)$ with $\Omega_R = |\langle b|\mathbf{d}\cdot\mathbf{E}_0|a\rangle|/\hbar$ (we can choose phases so Ω_R is real).

Expand $|\psi(t)\rangle = c_a(t)|a\rangle + c_b(t)e^{-i\omega_0 t}|b\rangle$. The Schrödinger equation gives

$$i\dot{c}_a = \frac{\Omega_R}{2} c_b \left[e^{i\phi}e^{i(\omega-\omega_0)t} + e^{-i\phi}e^{-i(\omega+\omega_0)t}\right],$$
$$i\dot{c}_b = \frac{\Omega_R}{2} c_a \left[e^{i\phi}e^{i(\omega+\omega_0)t} + e^{-i\phi}e^{-i(\omega-\omega_0)t}\right].$$

64.1 Rotating Wave Approximation

Dropping fast-oscillating terms ($\delta = \omega - \omega_0$):

$$i\frac{d}{dt}\begin{pmatrix} c_a \\ c_b \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 0 & \Omega_R e^{-i\phi}e^{-i\delta t} \\ \Omega_R e^{i\phi}e^{i\delta t} & 0 \end{pmatrix}\begin{pmatrix} c_a \\ c_b \end{pmatrix}.$$

64.2 Effective Hamiltonian

$$H_{\text{eff}} = \frac{\Omega_R}{2}\left[\hat{\sigma}_x \cos(\delta t + \phi) - \hat{\sigma}_y \sin(\delta t + \phi)\right].$$

64.3 Frame Change

Transform to the rotating frame via $U(t) = \exp[i(\delta t + \phi)\sigma_z/2]$:

$$H' = \frac{\Omega_R}{2}\sigma_x - \frac{\delta}{2}\sigma_z = \frac{1}{2}(\mathbf{\Omega}_{\text{eff}}\cdot\hat{\boldsymbol{\sigma}}), \quad \mathbf{\Omega}_{\text{eff}} = (\Omega_R, 0, -\delta).$$

64.4 Eigenvalues and Generalized Rabi Frequency

$$E = \pm\frac{1}{2}\sqrt{\Omega_R^2 + \delta^2}, \qquad \Omega = \sqrt{\Omega_R^2 + \delta^2}.$$

64.5 Time Evolution and Rabi Oscillations

Using $e^{i\alpha(\hat{n}\cdot\hat{\boldsymbol{\sigma}})} = I\cos\alpha + i(\hat{n}\cdot\hat{\boldsymbol{\sigma}})\sin\alpha$ with initial condition $c_a(0) = 1$:

$$c_a(t) = \cos\left(\frac{\Omega t}{2}\right) + i\frac{\delta}{\Omega}\sin\left(\frac{\Omega t}{2}\right),$$
$$c_b(t) = -i\frac{\Omega_R}{\Omega}\sin\left(\frac{\Omega t}{2}\right).$$

Rabi Oscillations

$$P_b(t) = \left(\frac{\Omega_R}{\Omega}\right)^2 \sin^2\left(\frac{\Omega t}{2}\right).$$

π pulse: complete population transfer. $\pi/2$ pulse: equal superposition.

65 Optical Bloch Equations

65.1 Density Matrix

$$\hat{\rho} = \sum_i p_i|\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad \mathrm{Tr}(\hat{\rho}) = 1.$$

For a two-level atom in basis $\{|e\rangle, |g\rangle\}$:

$$\hat{\rho} = \begin{pmatrix} \rho_{ee} & \rho_{eg} \\ \rho_{ge} & \rho_{gg} \end{pmatrix}, \quad \rho_{ee} + \rho_{gg} = 1, \quad \rho_{eg} = \rho_{ge}^*.$$

Define the Bloch vector:

$$u + iv = 2\rho_{eg}, \quad w = \rho_{ee} - \rho_{gg}, \quad \mathbf{r} = (u, v, w)^T.$$

Physical insight

The density matrix is the natural framework for open quantum systems: it handles statistical mixtures and allows dissipation to be incorporated via the Lindblad master equation.

65.2 Von Neumann Equation

$$\frac{d\hat{\rho}}{dt} = \frac{1}{i\hbar}[\hat{H}, \hat{\rho}], \quad \hat{H} = -\frac{\hbar\delta}{2}\sigma_z + \frac{\hbar\Omega_R}{2}\sigma_x.$$

Matrix elements:

$$\dot{\rho}_{ee} = \frac{i\Omega_R}{2}(\rho_{ge} - \rho_{eg}),$$
$$\dot{\rho}_{eg} = i\left[\delta\rho_{eg} - \frac{\Omega_R}{2}(\rho_{ee} - \rho_{gg})\right].$$

65.3 Dissipation Terms

$$\dot{\rho}_{ee}\big|_{\text{diss}} = -\frac{\rho_{ee}}{T_1}, \quad \dot{\rho}_{eg}\big|_{\text{diss}} = -\frac{\rho_{eg}}{T_2}, \quad \frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\phi.$$

65.4 Bloch Equations

$$\begin{aligned}\dot{u} &= -\delta v - u/T_2,\\ \dot{v} &= \delta u - v/T_2 + \Omega_R w,\\ \dot{w} &= -(w - w_0)/T_1 - \Omega_R v, \quad w_0 = -1.\end{aligned}$$

Bloch equations

$d\mathbf{r}/dt = \mathbf{\Omega} \times \mathbf{r} - \text{damp}$

Compact form:

$$\frac{d\mathbf{r}}{dt} = \mathbf{\Omega} \times \mathbf{r} - \begin{pmatrix} u/T_2 \\ v/T_2 \\ (w - w_0)/T_1 \end{pmatrix}, \quad \mathbf{\Omega} = (-\Omega_R, 0, \delta).$$

65.5 Example: Resonant Drive

For $\delta = 0$, no decay: $\mathbf{r}(t) = (0, -\sin(\Omega_R t), -\cos(\Omega_R t))$ — pure Rabi oscillation.

65.6 Steady-State Solution

Setting $\dot{u} = \dot{v} = \dot{w} = 0$ with saturation parameter $s = \Omega_R^2 T_1 T_2$:

$$\begin{aligned} u_{\text{ss}} &= \frac{\delta \Omega_R T_2^2}{1 + \delta^2 T_2^2 + s}, \quad w_{\text{ss}} = -\frac{1 + \delta^2 T_2^2}{1 + \delta^2 T_2^2 + s},\\ \rho_{ee}^{(\text{ss})} &= \frac{s/2}{1 + \delta^2 T_2^2 + s}.\end{aligned}$$

As $s \to \infty$: $\rho_{ee} \to 1/2$ — populations equalize but never invert.

66 Jaynes–Cummings Model

The JC model is the paradigm of cavity QED and circuit QED. It predicts vacuum Rabi splitting, photon-number-dependent dynamics, and collapse and revival of Rabi oscillations.

66.1 Atomic Hamiltonian

For a two-level atom with states $|e\rangle$, $|g\rangle$, ignoring the constant average energy:

$$H_{\text{atom}} = \frac{\hbar \omega_a}{2} \sigma_z, \quad \sigma_z = |e\rangle\langle e| - |g\rangle\langle g|.$$

Raising/lowering operators: $\sigma_+ = |e\rangle\langle g|$, $\sigma_- = |g\rangle\langle e|$.

66.2 Field Hamiltonian

$$H_{\text{field}} = \hbar \omega_c a^\dagger a.$$

66.3 Interaction Hamiltonian and JC Model

The dipole interaction $H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}$ with $E = \epsilon E_0 (a + a^\dagger)$ and $d = d_{eg}(\sigma_+ + \sigma_-)$ gives

$$H_{\text{int}} = \hbar g (\sigma_+ a + \sigma_- a + \sigma_+ a^\dagger + \sigma_- a^\dagger), \quad g = -\frac{d_{eg} \epsilon E_0}{\hbar}.$$

In the interaction picture, with $a(t) \sim e^{-i\omega_c t}$, $\sigma_+(t) \sim e^{i\omega_a t}$,

$$H_I(t) = \hbar g\Big(\sigma_+ a\, e^{i(\omega_a-\omega_c)t} + \sigma_- a^\dagger e^{-i(\omega_a-\omega_c)t} + \underbrace{\sigma_+ a^\dagger e^{i(\omega_a+\omega_c)t} + \sigma_- a\, e^{-i(\omega_a+\omega_c)t}}_{\text{fast, drop via RWA}}\Big).$$

Defining detuning $\delta = \omega_a - \omega_c$, the Jaynes–Cummings Hamiltonian (in RWA) is:

$$\boxed{H_I(t) = \hbar g\left(\sigma_+ a\, e^{i\delta t} + \sigma_- a^\dagger e^{-i\delta t}\right).}$$

66.4 State Evolution

Within each two-dimensional manifold $\{|e,n\rangle, |g,n+1\rangle\}$:

$$|\psi(t)\rangle = C_e(t)|e,n\rangle + C_g(t)|g,n+1\rangle.$$

The equations of motion are

$$\dot{C}_e(t) = -ig\sqrt{n+1}\, e^{i\delta t} C_g(t),$$
$$\dot{C}_g(t) = -ig\sqrt{n+1}\, e^{-i\delta t} C_e(t).$$

66.5 Resonance Case ($\delta = 0$)

Solution: $\ddot{C}_e = -g^2(n+1)C_e$, giving

$$C_e(t) = A\cos\left(g\sqrt{n+1}\, t\right) + B\sin\left(g\sqrt{n+1}\, t\right).$$

Case 1: $C_g(0) = 1$, $C_e(0) = 0$:

$$C_e(t) = -i\sin\left(g\sqrt{n+1}\, t\right), \quad C_g(t) = \cos\left(g\sqrt{n+1}\, t\right),$$
$$P_{g,n+1} = |C_g|^2 = \tfrac{1}{2}(1+\cos\Omega_n t).$$

Case 2: $C_e(0) = 1$, $C_g(0) = 0$:

$$C_e(t) = \cos\left(g\sqrt{n+1}\, t\right), \quad C_g(t) = -i\sin\left(g\sqrt{n+1}\, t\right),$$
$$P_{e,n} = |C_e|^2 = \tfrac{1}{2}(1+\cos\Omega_n t).$$

Jaynes–Cummings Rabi Frequencies

$\Omega_n = 2g\sqrt{n+1}$ (resonance), $\Omega_n = \sqrt{\delta^2 + 4g^2(n+1)}$ (off resonance).

For $n = 0$ (vacuum Rabi oscillation): $\Omega_0 = 2g$. This is a purely quantum effect with no classical analogue.

Physical insight

The $\sqrt{n+1}$ photon-number dependence of Ω_n causes collapse and revival of atomic inversion when the field starts in a coherent state — a landmark quantum phenomenon first observed by

These notes cover the theoretical foundations of quantum optics:

1. **Quantization**: field modes as QHOs; Fock, coherent, and squeezed states; quadratures; symplectic structure.
2. **Nonlinear optics**: $\chi^{(2)}$ interactions; SPDC; two-mode squeezing; polarization entanglement.
3. **SU(2)/SU(1,1)**: Schwinger representation; beam splitters; MZI; SQL; HOM effect; N-photon states.
4. **Measurement**: Balanced homodyne detection; mode mismatch; optical attenuation; multimode theory.
5. **QKD**: BB84 protocol; eavesdropping detection; two-photon improvement.
6. **Atom–light**: Dipole Hamiltonian; Göppert-Mayer transformation; Fermi's golden rule; Rabi oscillations; Bloch equations; Jaynes–Cummings model; vacuum Rabi oscillation.